InventOR: Can Large Language Models Set Reorder Points and Safety Stocks? Industrial Evidence Against ERP and Data-Driven Benchmarks

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Abstract

Can a large language model (LLM) revise material-level replenishment parameters effectively? We evaluate a prompt-configured LLM, without project-specific training, inside *InventOR*: it proposes a reorder-point/order-quantity (r, Q) policy with a justification; deterministic software checks and replays it; a planner retains decision authority. For 365 plant-material pairs from three plants of one manufacturer, three original LLM runs, seven reference policies and an ablation withholding ERP planning parameters are evaluated on held-out demand. Three further generations on a masked public release test replication, not six independent datasets. On the original 346-pair common-feasible cohort, the LLM costs about 11% less than ERP-derived parameters but 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile baselines. Policy summaries resemble the prompt's textbook recipe, with median smoothed safety-stock and order-quantity ratios of 1.00 and reorder-point ratio 1.09. The mean within-pair cost range across original runs is 10.5% of pair mean cost. Code using ERP safety lead time costs 44% more than the LLM. Rescoring fixed policies with lost sales priced at 1% of unit cost or more largely compresses non-tuned cost differences. Unconstrained trajectories breach storage limits for 53–66% of pairs; capacity-clipped replay can violate order-level MOQ. Masked runs preserve the principal ranking.

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InventOR supports checkable candidate generation, not demonstrated optimization superiority or improved planner outcomes.
Keywords: Inventory Control, Large Language Models, Generative Artificial Intelligence, Safety Stock, Intermittent Demand, Empirical Evaluation

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point, a safety stock and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand, supplier performance and storage constraints change. Revisiting them requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. A conversational large language model (LLM) with a code interpreter appears to offer this scarce skill, and a growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5, 6], building on the long-standing promise of data analytics in logistics [7].

Three problems limit its use. First, LLM outputs can be unfaithful or fabricated [8], so a proposed policy is a candidate, not a decision. Second, superiority claims over ERP parameters are hard to interpret when the LLM and the comparators receive different information. Third, a storage rule that holds when a policy is written down can be violated as stock builds up over time.

Prior LLM work includes solver-code generation, inventory-control agents and human–AI simulation optimization (Section 3). Our focus is narrower: fixed per-material (r, Q) proposals for an industrial portfolio, evaluated against ERP-derived parameters and seven reference arms with explicit input boundaries. We call the workflow *InventOR*: for each plant-material pair, an enterprise LLM application receives a fixed prompt, with no project-

specific training or fine-tuning, and proposes a candidate (r, Q) policy with a justification and escalation flag. Deterministic software parses, gates and simulates proposals; a planner retains authority over ERP changes (Section 2).

We study five questions on real industrial data: what cost, service and inventory the LLM policies achieve relative to ERP-derived parameters, data-driven baselines and textbook code, with explicit input boundaries (RQ1); how their differences relate descriptively to parameter similarity, run-to-run noise, demand segments and shortage prices (RQ2); whether policies respect storage limits along simulated trajectories (RQ3); how stable they are across repeated generations (RQ4); and which plant and demand-regime results might inform proposed review and escalation rules (RQ5).

We contribute (C1) a workflow with explicit requirements that bound the LLM's role (Sections 2 and 4); (C2) an evaluation with seven reference arms, code implementing the prompt's formulas, an ablation withholding ERP planning parameters and hard-limit replay (Section 5); (C3) three original runs on 365 target pairs and three fresh generations on a masked version of the same panel, with a matched six-run report, cost decomposition, shortage-price break-even and descriptive mechanism analyses (Section 6); and (C4) implications for practice and research (Section 7).

Against ERP-derived parameters the original LLM policies cost about 11% less at mean fill 0.1–0.4 points lower, but cost 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile policies at similar mean fill. This is not an equal-service experiment. Static-compliant policies exceed storage limits on unconstrained trajectories for 53–66% of pairs. Smoothed median policy ratios resemble textbook code, with reorder points 9% higher; these summaries do not reveal internal calculations. Code using ERP inputs costs 44–45% more than the LLM, illustrating the importance of input treatment. Pricing lost sales at 1%

of unit cost or more brings five policy arms within 4% of each other; the simulation-tuned arm is cheaper under these fixed-policy revaluations. The six-run matched report preserves the main ranking. Written rationales and flags are possible review aids, not measured planner benefits or development-effort savings.

2. Problem Setting and Workflow

2.1. The decision problem

The decision supported is the periodic (re)setting of a reorder point, a safety stock and an order quantity for each plant-material pair held in an ERP system. It is a partly structured operational-control decision [9]: the calculation is textbook, but the choice of estimator, the treatment of intermittent demand, the plausibility of recorded lead times and the handling of constraints require judgement that is applied inconsistently when thousands of materials compete for a few analysts' time. The decision-maker is the material planner; hard constraints are the minimum order quantity (MOQ) and a physical storage limit; and feedback is slow, because a changed parameter shows its effect only in later demand. The scope is one pair at a time (single item, single echelon, continuous review, lost sales, (r, Q) family); order release, supplier selection, forecasting-model development and multi-echelon networks are out of scope, and the LLM is a candidate generator that does not forecast, place orders, simulate or decide.

2.2. The InventOR workflow and its level of automation

> **Definition.** *InventOR is a human-reviewed planning workflow that, for one plant-material pair at a time, (i) obtains from a large language model, configured only by a written prompt, a candidate (r, Q) policy with a justification and an*

escalation flag; (ii) verifies the candidate deterministically against explicit constraints; (iii) evaluates it by counterfactual simulation against reference policies on held-out demand; and (iv) presents the result to a planner, who alone decides whether ERP parameters are changed.

In the levels of automation of Parasuraman et al. [10], InventOR operates at a low-to-intermediate level: it suggests one alternative and leaves execution to the human. Accepting suggestions invites complacency and automation bias [11], which the review rules of Section 4.3 are meant to counter.

2.3. Requirements on the workflow

The requirements follow from the known unreliability of LLM outputs [8], guidance on human–AI interaction and accountability [12, 13], run-to-run nondeterminism of LLM inference [14] and the constraints of inventory control. Table 1 ties each to a design feature (Section 4) and an evaluation (Section 6).

Table 1: Requirements on the workflow, the features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.10
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.8
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

3. Related Work and Positioning

Human–AI decision making in planning.. Decision support aims to improve the judgement of a human decision-maker, not to replace it [9, 15, 16], and how suggestions are scoped, presented and verified determines whether users rely on them appropriately [10, 11]. In inventory control, human override of algorithmic safety-stock advice is a documented risk [17], people discount algorithms after seeing them err [18], and AI-based route planning yields efficiency advantages when it is followed in full [19]. InventOR therefore keeps the planner as decision-maker and makes the evidence for each proposal explicit.

LLMs in operations and supply-chain decisions.. Wu [20] compares automated and human-specified LLM-assisted inventory optimization under supply-chain disruptions through numerical experiments. Our study instead freezes per-material policies for held-out replay on an industrial portfolio; it is not the first use of LLMs in inventory optimization. Reviews and frameworks map the opportunities of generative AI in supply chains [1, 2, 3, 21, 4, 5] and stress human roles, data quality and governance [6, 22, 23]. Evidence from manufacturing planning is thin: a fine-tuned LLM for digital-manufacturing optimization [24], a transformer trained to set order timing and quantity [25] and evidence that ChatGPT reproduces managers' decision biases [26]. Most technical LLM work translates problem statements into solver code [27, 28, 29, 30] or studies multi-agent control in inventory flows [31, 32]. The closest conceptual neighbour studies how OR algorithms, LLM agents and humans complement each other in multi-period inventory control on a benchmark of more than 1,000 instances, and reports that OR-augmented LLM methods outperform either alone [33]; we ask the prior question of how far an LLM alone gets on industrial data. In forecasting, LLMs do not consistently outperform human judgment in retail [34], and an LLM forecasting copilot for spare-parts demand shows high variability for irregular demand that averaging repeated

queries reduces [35]. We test run-to-run variability directly (Section 6.8) and let the LLM propose a complete policy, not a forecast.

Data-driven inventory control without LLMs.. Learning-based replenishment includes policies learned from non-stationary demand [36], forecasts with leading indicators [37], deep reinforcement learning for omnichannel and multi-echelon systems [38, 39, 40], contextual robust optimization [41] and reinforcement learning combined with demand forecasting [42]; heuristics remain competitive in constrained settings [43]. A trained framework jointly sets replenishment parameters, safety stock and safety lead time on automotive data [44]. Forecasting research links forecasts to safety stock [45], combined-forecast error distributions [46] and inventory-cost objectives [47]. In *Expert Systems with Applications*, model-based deep reinforcement learning addresses new products with short histories [48], and contextual agents adapt to unseen supply-chain configurations [49]. TFT-MPIR directly generates multi-period replenishment orders from historical and contextual inputs [50]; graph recurrent networks address store-SKU replenishment [51]; RevIN-TSMixer connects demand forecasting to continuous-review inventory costs [52]. InventOR instead evaluates a fixed-prompt, pretrained generator without project-specific model fitting; it does not establish superiority over these trained methods.

Intermittent and spare-parts demand.. Manufacturing histories are often intermittent or lumpy. Work began with the bias of Croston's estimator and its stock-control performance [53, 54] and continues with neural [55], bootstrapping [56], obsolescence-aware [57], temporal-hierarchy [58] and state-space [59] forecasts, order-up-to control [60], safety-stock placement [61] and lead-time forecasting [62], all of which bear on the inputs of the policies we compare. The wider literature adds specialized estimators and classification schemes [63, 64, 65, 66, 67], spare-parts forecasting [68] and multi-echelon spare-parts models [69]. InventOR is not a

new inventory model and does not claim to beat trained or optimized methods; its (r, Q) policy is the standard workhorse of continuous-review control [70, 71, 72].

4. The InventOR Workflow

4.1. Architecture and components

Following the classical decision-support components [15, 16], InventOR has a *data* component (a material-scoped extract of daily demand, lead times, working-day flags, cost and constraint parameters and ERP reference values; the fields supplied are controlled and logged; DR4), a *knowledge* component (the system prompt file defining the task, the prohibited methods, the storage rules and the output schema, Supplementary File S1; DR1, DR5), a *model* component split into a generator (an enterprise LLM application with a code interpreter, configured only by the prompt, proposing one (r, Q) policy with a justification and an escalation flag; DR1) and a deterministic verifier and simulator (parser, gate, held-out simulation with hard storage-limit replay and reference arms; DR2, DR3), a *dialog* component (policy, justification, flag and simulated comparison shown to the planner under the trust, escalate and reject rules; DR6) and a *governance* layer (frozen prompt file, per-call log of input hash, time, session and raw response, repeated runs; DR5). The generator is the only non-inspectable component. In the evaluated prototype the dialog component is the per-pair output record; no interactive interface was built or evaluated.

4.2. Operation in four steps

For one material, InventOR answers how much safety stock to hold, at what level to reorder and how much to order, in four steps (Figure 1).

Step 1. Generate. The LLM, accessed through an enterprise application with a code interpreter, receives one material's history (daily demand, realized lead times, working-

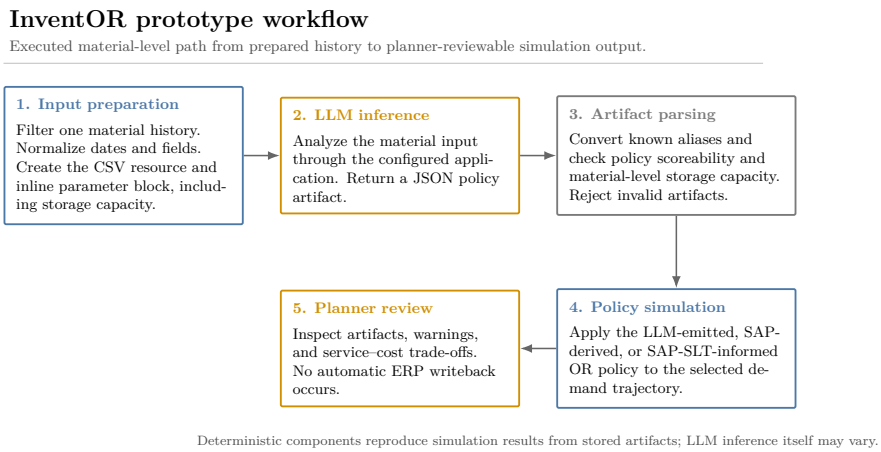


Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

day flags) and a short parameter block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [73] instructs it to compute demand and lead-time statistics, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material's demand observed *after* the data cutoff, a held-out evaluation in the sense of out-of-sample forecast testing [74, 75], together with the comparator policies, producing holding and or-

dering cost, fill rate and stockout days.

Step 4. Review. A planner sees the policy, the LLM's justification and flag, and the simulated comparison. Only the planner changes ERP parameters. The planner's decision (accept, adjust or reject), the review time and an optional free-text comment are logged with the call, so that practitioner feedback accumulates as a by-product of use; the prototype evaluated here does not yet implement this logging.

The LLM is a *candidate generator* whose output is never trusted without the gate; it does not forecast, place orders or run the simulation. The prompt prohibits Monte Carlo simulation, distribution fitting, machine-learning libraries and specialized intermittent-demand estimators, so that the policy rests on transparent statistics.

4.3. Trust, review and reject rules

We propose three rules for planners, in line with guidelines for human–AI interaction and AI auditing [12, 13]. The rules and the 0.5-point threshold are design proposals not evaluated with planners; the trust rule would be met by a minority of pairs (the LLM is cheaper than the history-only OR in 74–77 of 346).

Trust (consider after review) only after checking both order-level MOQ and storage compliance, as well as a simulated cost no higher than the best history-only reference (Croston/SBA or empirical quantile) with fill not more than about 0.5 points lower. Capacity-clipped replay alone cannot establish the joint compliance needed to apply this rule.

Escalate when the LLM raises an escalation flag, the material is lumpy or has strongly seasonal or trending demand, the plant or calendar data were repaired, or the policy differs strongly between repeated generations.

Reject when the proposal fails the gate after the retry budget, exceeds the storage limit in the hard-limit replay, or is dominated (higher cost and lower fill) by a history-only reference.

4.4. Provenance and reproducibility of the configuration

The system prompt, a plain-text file deployed unchanged in the enterprise application, defines the task, forbidden methods, capacity rules and output schema (Supplementary File S1). Every call logs the SHA-256 of its input, UTC time, session identifier and raw response. The platform does not expose the model identifier or sampling settings; the evaluated model (Anthropic Claude Sonnet 4.6 with a code interpreter) is operator-declared. Reproducibility rests on the prompt file, the call logs and three independent runs (Section 6.8).

5. Application and Evaluation Design

5.1. Evaluation approach

The evaluation addresses the requirements of Table 1. The primary basis is the hard-limit replay with working-day receipts and no shortage penalty; sensitivity analyses vary these choices. This is an ex post simulation study with real demand, simplified dynamics and no live users. The history-only, SLT-informed and ERP-floor OR arms implement the prompt's EOQ and normal-approximation recipe, with different treatments of ERP inputs. The experiment characterizes policy cost, service and constraint failures, not usability, adoption or improved planner decisions.

5.2. Industrial application and data

The data come from a large manufacturer, and the evaluation is retrospective. The dataset contains 233,078 daily records for 411 plant-material pairs from three plants (A, B and C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed to 27 August

2026. Of the 411 pairs, 365 are eligible for scoring (234 in Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19 pairs (all in Plant A) the MOQ plus the reference safety stock exceeds the storage limit, so no comparator can satisfy both rules; these are excluded from outcome comparison, leaving a *common-feasible cohort* of 346 pairs (215, 65 and 66). Demand is concentrated (the five largest materials account for 40.3% of demand). The source working-day flag of Plant C is all zero; we reconstruct it from the same-date majority of Plants A and B and test this repair in Section 6.10. Data quality is a known threat to decision support [76] and inventory records [77, 78, 79].

Each pair is sent as a material-scoped CSV with daily demand, recorded lead times, a working-day flag, the minimum order quantity, unit cost, holding-cost rate, ordering cost, service-level target, the storage limit, and two *ERP planning parameters* (current safety stock and safety lead time), which are withheld in the ablation arm. The CSV also carries supplier and lane context fields that the prompt does not use. A masked version of the panel is released for replication (Section 6.9); there the plants are labelled PL1, PL2 and PL3 (corresponding to A, B and C).

5.3. Reference arms and information content

ERP safety stock and safety lead time embed planner judgement and are alternative ways of buffering unreliability [80], and planned lead times can drift upward [81]. The LLM and the classical comparators therefore do not necessarily use the same information, and we construct arms that make it explicit (Table 2).

Three further arms use no ERP planning inputs. *Croston/SBA OR* uses bias-corrected Croston demand [63, 64], smoothing 0.1, and root mean squared one-step forecast error as demand spread [65]. *Empirical-quantile OR* uses the service-target quantile of rolling

Table 2: Comparison arms and their information. Policy quantities satisfy MOQ and are projected to the storage limit when possible; capacity-clipped orders in replay can fall below MOQ.

Arm	Parameters from	ERP inputs used	Purpose
SAP-derived	ERP safety stock and safety lead time; EOQ order quantity	Both	ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time	OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both	OR arm matched to the LLM's full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None	ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None	Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None	Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None	Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both	Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to "not supplied"	None	Ablation: what the LLM does when it cannot anchor on ERP

demand over rounded mean lead time. *Simulation-tuned* (r, Q) searches reserve factors $-0.5, -0.25, \ldots, 3.5$ and quantity multiples $0.5, 0.75, 1, 1.5, 2$ (up to 85 candidates), selecting the cheapest meeting the service target, or the best fill if none meets it. Statistics use all pre-cutoff data; tuning replays the last 250 working days without capacity clipping, initialized at each candidate's $R + Q$ with an empty pipeline. This is in-sample tuning, not nested validation; evaluation instead uses shared observed opening stock and capacity clipping. No post-cutoff demand informs selection. Deep reinforcement learning [39, 38] is not benchmarked.

5.4. Simulation, capacity replay and metrics

Each pair is replayed as a single-item, single-echelon (r, Q) system [70, 82, 83, 72, 71] with lost sales [84], a discrete working-day approximation to continuous review, shared observed opening stock, an empty pipeline and deterministic arrivals after the rounded mean lead time (Appendix Appendix A). This is a simulation experiment, not a reconstruction of actual orders [85]. The policy quantity satisfies the supplier MOQ [86], but the hard-limit replay prioritizes capacity [87]: clipped orders may fall below MOQ. Metrics are mean material-horizon holding-plus-ordering cost, mean material-level and volume-weighted fill rates, and stockout days. Paired 10,000-resample bootstrap 95% intervals describe this cohort; material dependence within three plants precludes population-level inference.

A static check $SS + Q \leq$ `max_storage_units` does not show that storage limits hold as inventory evolves, because the reorder point includes lead-time demand and inventory on order can exceed the limit on arrival. We therefore record the post-receipt on-hand stock for every policy and day and report the share of pairs and days above the limit (the *unconstrained* simulation), and we *replay* every policy with each order truncated to the storage limit minus the current inventory position (the *hard-limit replay*), which is our primary evaluation.

5.5. Repeated runs and sensitivities

To examine run-to-run variation, the entire cohort is generated three times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1 serves as the reference for tabulated results unless stated otherwise; runs 2 and 3 provide replicates. Sensitivity analyses vary the working-day reconstruction (majority, any-plant, strict-all, exclusion of Plant C and calendar-day arrivals), the shortage penalty (zero, one and five times unit cost).

6. Results

6.1. Completion and policy characteristics

All four runs (three with the full input and one ablation, Section 6.4) of the original data return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most three attempts per pair. The validity gate rejects non-positive reorder points and order quantities, negative safety stocks and reorder points below the safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of the 346 in the cohort below), so for a majority of pairs the SAP-derived arm holds no safety stock at all.

6.2. Headline comparison with ERP-derived parameters

Unless stated otherwise, all results use the hard-limit replay on the 346-pair common-feasible cohort with working-day receipts and no shortage penalty (Section 6.5 gives the unconstrained simulation). Table 3 compares all arms; its last columns give the LLM cost relative to each arm in the three runs (positive: the LLM costs more). In run 1 the LLM policies cost 11.4% less than SAP-derived policies and 30.5% less than the SLT-informed OR (mean 2,928 versus 3,305 and 4,211), at a mean fill 0.18 points lower than SAP-derived (98.11% versus 98.30%; interval -0.54 to $+0.10$). The mean cost difference against SAP-derived is -377 (95% interval -651 to -154); the LLM is cheaper in 241 of 346 pairs and, in 226, simultaneously no more expensive and no worse in fill. It incurs more stockout days (910 versus 804). The volume-weighted (aggregate) fill rate is only 78.7–79.3% for every arm except the simulation-tuned policy (84.8%), against a mean material-level fill of about 98%: with lost sales unpriced, service on high-volume materials is low under all policies, so the cost ranking is not a comparison at equal service.

Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and about 16% lower holding cost at about the same number of

orders per year (17.3 versus 17.6), and it accepts 11–20% more stockout days (runs 3 and 2). The saving is therefore a trade of inventory for service, not a better ordering rhythm, and Section 6.10 shows that once shortages carry a cost this trade no longer favours the LLM.

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year. The last column gives the LLM cost relative to the arm in runs 1/2/3 (positive: the LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	−11.4 / −11.3 / −11.4
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	−30.5 / −30.4 / −30.5
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	−31.1 / −31.0 / −31.1
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	+5.9 / +6.0 / +5.9
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	+8.2 / +8.4 / +8.3
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	+17.7 / +17.8 / +17.7
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	−10.4 / −10.3 / −10.3‡
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	−8.8† (vs SAP)

† Other comparisons of the ablation are in Section 6.4. ‡ Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.10).

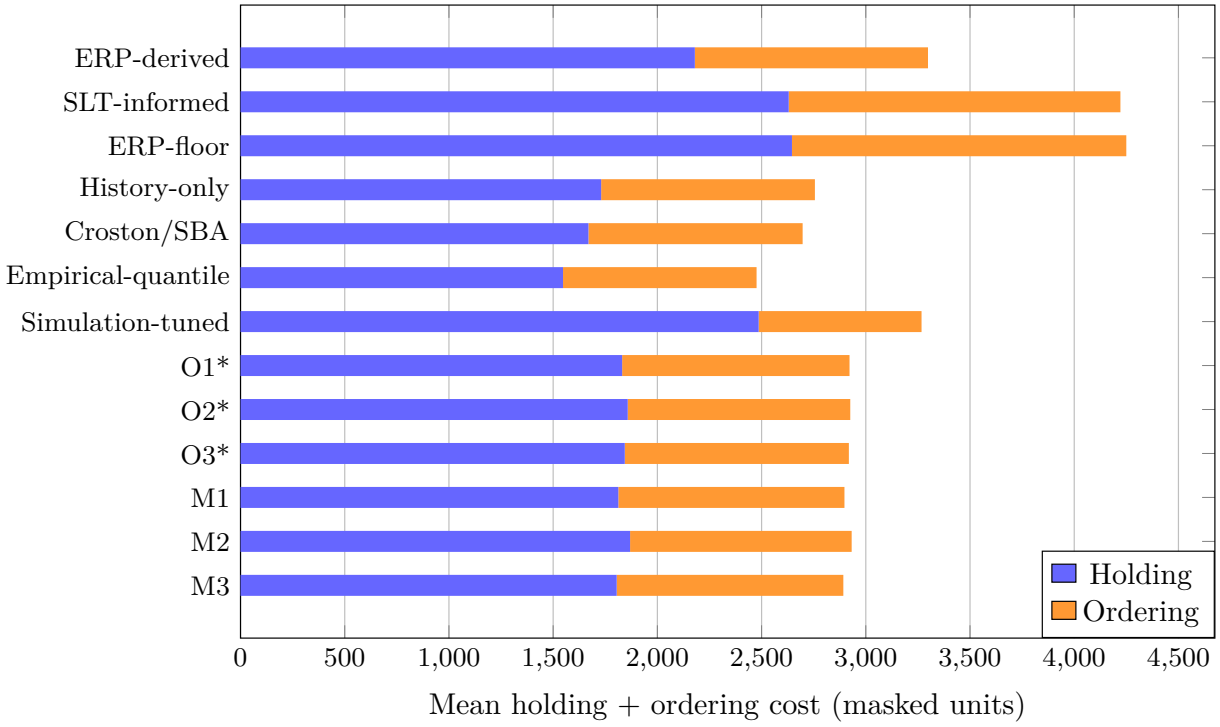


Figure 2: Six-run cost decomposition on the same 338 feasible masked-panel pairs (zero shortage penalty). O1*–O3* are frozen original policies rescaled to the masked panel, not fresh inference; M1–M3 are new masked generations. Comparator policies are identical across these replays. Holding and ordering costs use masked units; they are not the original-scale values in Table 3.

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248): the ERP safety lead time inflates their effective lead time and buffers. Against them the LLM is 30–31% cheaper in every run, which mostly measures how heavily ERP inputs over-buffer.

Second, the *history-only OR*, which uses no ERP inputs, costs 2,764 at a fill of 98.14%. The LLM costs 5.9–6.0% more in all three runs (run 1: +164, interval +91 to +240), with fill differences of −0.03, −0.25 and +0.01 points whose intervals include zero, and is cheaper in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap: the LLM costs 8.2–8.4% more than the Croston/SBA OR (2,705, fill 98.30%; run 1: +223, interval +97 to +362; cheaper in 91–99 pairs) and 17.7–17.8% more than the empirical-quantile OR (2,489, fill 98.35%; +439, interval +329 to +557; cheaper in 44–49 pairs; mean fill 0.24 points lower). Part of the LLM's advantage over ERP-derived parameters therefore reflects weak ERP benchmarks.

The simulation-tuned (r, Q) policy sits at a different operating point: an aggregate fill of 84.8% and the lowest inventory (11,914 units) but a mean fill of 96.05% and 1,973 stockout days, because it is tuned on the pre-cutoff year. The LLM costs 10.3–10.4% less at zero penalty with a 2.1-point higher mean fill, but the interval includes zero (−2,527 to +857); once shortages are priced the tuned policy is by far the cheapest arm (Section 6.10).

6.4. LLM without ERP planning parameters (ablation)

This ablation blanks ERP safety stock and safety lead time, marking them "not supplied", with the same prompt and other inputs. All 365 original pairs returned a static-compliant policy. Median parameter ratios to run 1 are 1.00, but reorder points differ by more than 10% in 54.6% of pairs, against 48.3% and 48.0% between full-input runs. Hard-limit cost

is 3,014 at mean fill 98.06%: 8.8% below SAP-derived (interval −566 to −61), but 9.0% above History-only (interval +134 to +387), 11.4% above Croston/SBA and 21.1% above empirical-quantile. Relative to SAP-derived, costs are −10.8%, −21.9% and +41.6% in Plants A/B/C. The Plant C reversal therefore persists without these fields; one stochastic ablation does not isolate their causal effect or prove that the LLM ignores them. Costs, MOQ, service target and capacity remain available to all arms.

6.5. Storage limits along simulated trajectories

All 365 original LLM policies satisfy $SS + Q \leq$ `max_storage_units`, yet unconstrained post-receipt stock exceeds the limit for 183 pairs (52.9%; 178–183 across runs; 10.2% of pair-days) for the LLM, 198 (57.2%) for SAP-derived and 230 (66.5%) for SLT-informed OR. The static rule does not bound pipeline accumulation and is insufficient for trajectory feasibility. In the hard-limit replay no arm exceeds the limit by more than one unit (integer truncation of inventory position); this is not simultaneous proof of hard MOQ compliance. Enforcing capacity raises LLM mean cost from 2,850 to 2,928 and SAP-derived from 3,184 to 3,305 without changing the ranking.

6.6. Plants and demand regimes

The average hides a strong plant effect. In original run 1 the LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of 215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant C (cheaper in 8 of 66); the Plant C loss is similar in the other full runs (+32.2%, +31.5%) and persists under every working-day reconstruction in which Plant C can be scored (Section 6.10). Figure 3 separately shows all six runs on the matched masked-scale cohort. Against the data-driven baselines the original run 1 LLM costs more in every plant: 7.2%, 3.6% and 6.7% more than the history-only OR, 13.1%,

2.3% and 6.2% more than the Croston/SBA OR, and 20.8%, 7.7% and 29.5% more than the empirical-quantile OR in Plants A, B and C. The SLT-informed and history-only ORs coincide in Plant C because ERP safety lead time is zero for all its pairs.

By demand regime, the 33 lumpy materials show the largest reduction against SAP-derived (−25.4% in run 1), but the history-only and Croston/SBA ORs cost about 9–10% less than the LLM there; with 33 materials this is indicative only.

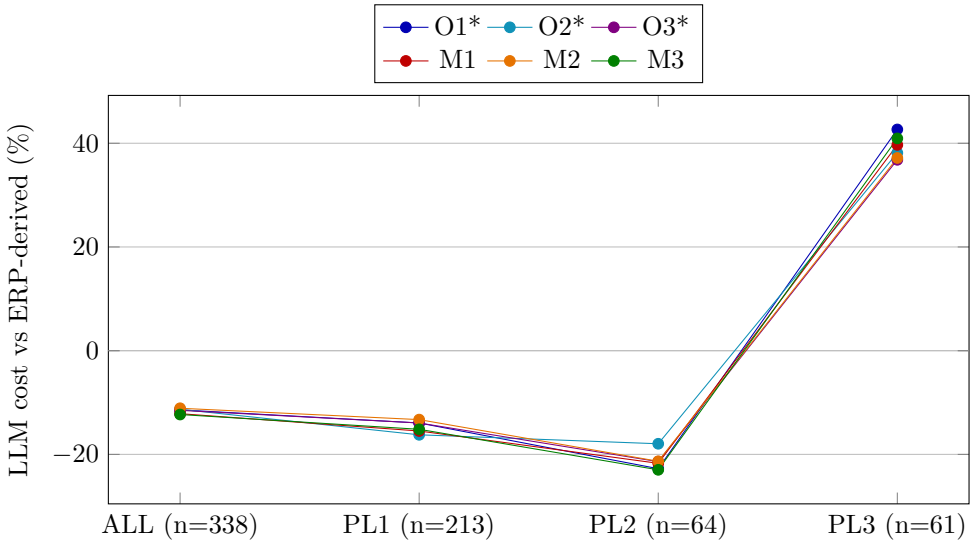


Figure 3: Plant heterogeneity across all six runs on a shared masked-panel cohort. Positive values mean the LLM costs more than ERP-derived parameters. O1*–O3* denote rescaled frozen original policies; M1–M3 denote fresh masked generations. PL1/PL2/PL3 correspond to Plants A/B/C. Lines connect descriptive group summaries, not time points or uncertainty intervals.

6.7. Descriptive comparison with formula-based code

The prompt prescribes an economic order quantity with a minimum-order-quantity floor and a normal-approximation safety stock (Appendix Appendix A). The History-only OR applies these formulas as code to the same demand and lead-time history, and the SLT-informed and ERP-floor ORs apply them with the ERP inputs the LLM also receives (Section 5). The comparison with code is therefore direct.

The LLM resembles the formula-based policies.. The median smoothed ratio $(x_{\text{LLM}} + 1)/(x_{\text{code}} + 1)$ is 1.00 for order quantity (medians 2,810 and 2,828 units) and safety stock (712 and 718, against an ERP median of 0); LLM parameters are averaged over the three original runs first. The corresponding reorder-point ratio is 1.09, including across the segments in Table 4, where LLM cost is 4–9% higher. These summaries indicate similarity and a reorder-point uplift, not item-level proof that the LLM executes the formulas faithfully.

Input treatment changes the ranking against code.. Code that receives the same ERP inputs as the LLM costs 4,211 (SLT-informed OR) and 4,248 (ERP-floor OR) per pair against 2,928 for the LLM in run 1, i.e. 44–45% more than the LLM, because the ERP safety lead times inflate the textbook reorder point. Code that ignores the ERP planning inputs costs 2,764, 5.9% less than the LLM. The LLM resembles the History-only code rather than the ERP-inflated code, which is consistent with the ablation in Section 6.4.

Run-to-run noise.. The within-pair cost range across the three original runs, divided by that pair's mean and averaged across pairs, is 10.5%; this is not a standard deviation. The runs disagree on beating History-only code for 34.1% of pairs. The LLM mean is cheaper than History-only for 17.1% of pairs and SAP-derived for 76.3%. Selecting each pair's best run with hindsight gives mean cost 2,765 (History-only: 2,764); selecting the worst gives 3,146. Such selection is not an implementable out-of-sample policy.

Plant C and segments.. Table 4 shows that the LLM advantage over SAP-derived parameters disappears in storage-bound pairs (+4.9% against −14.9% elsewhere) and reverses in Plant C (+33.3%); it does not vary monotonically with demand variability. Plant C is not an LLM-specific failure: the History-only code is also 28% more costly than SAP-derived parameters

there, so the ERP parameters are leaner than any formula-based policy in that plant when lost sales are not charged. Plant C also has the least regular demand (median coefficient of variation of daily demand 0.61, lead-time coefficient of variation 0.78, against 0.51 and 0.63 over all pairs).

Table 4: Where the LLM differs from code. Original-scale mean cost per pair in the hard-limit replay with zero shortage penalty (LLM: mean of three original runs; History-only: textbook code; SAP: ERP-derived), relative costs (%), and the median smoothed reorder-point ratio $(R_{\text{LLM}} + 1)/(R_{\text{code}} + 1)$. Storage-bound pairs require downward projection because textbook safety stock plus order quantity exceeds capacity.

Segment	n	LLM	History-only	SAP	vs History-only (%)	vs SAP (%)	ROP ratio
All pairs	346	2,929	2,764	3,305	+6.0	-11.4	1.09
Plant A	215	2,370	2,231	2,782	+6.2	-14.8	1.08
Plant B	65	5,050	4,746	6,360	+6.4	-20.6	1.10
Plant C	66	2,662	2,549	1,997	+4.4	+33.3	1.11
Demand CV, lowest third	115	2,249	2,092	2,707	+7.5	-16.9	1.09
Demand CV, middle third	115	3,872	3,659	4,152	+5.8	-6.8	1.10
Demand CV, highest third	116	2,669	2,544	3,057	+4.9	-12.7	1.08
ERP safety stock zero	188	2,946	2,741	3,215	+7.5	-8.4	1.09
ERP safety stock positive	158	2,909	2,792	3,411	+4.2	-14.7	1.09
Storage-bound	145	1,491	1,408	1,422	+5.9	+4.9	1.10
Not storage-bound	201	3,967	3,743	4,663	+6.0	-14.9	1.08

6.8. Run-to-run stability

Original aggregate costs relative to SAP-derived are −11.4%, −11.3% and −11.4%. Figure 4 extends the ranking comparison to all six runs on shared masked-panel pairs. Individual policies still vary: original runs 2 and 3 have identical triples for only 36 of 365 pairs. Similar aggregate means do not establish item-level repeatability [14].

6.9. Replication on the masked release

The raw data cannot be shared, so we publish a masked release of the panel (see Data and Code Availability): all quantities and prices are multiplied by secret per-pair factors, identifiers are replaced by pseudonyms, and dates and plant structure are unchanged. We repeated the generation three times on the masked panel with the unchanged prompt file (runs m1–m3), and scored the output with the same replay as above. Run m1 produced 364 and runs m2 and m3 all 365 of the 365 policies (one pair failed all three attempts in m1);

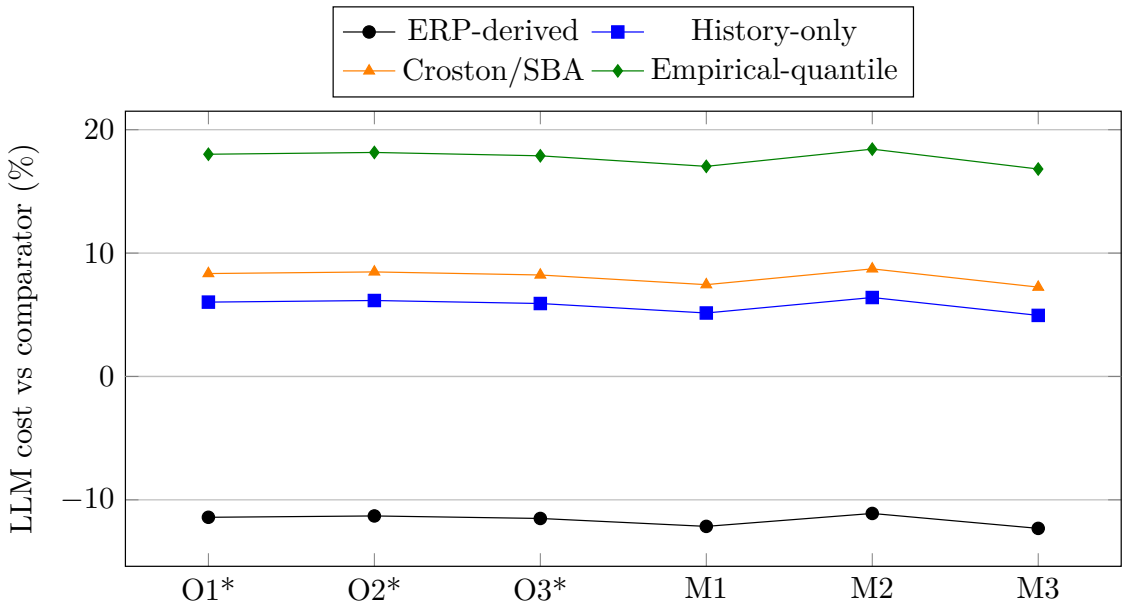


Figure 4: Ranking stability across six replays on 338 shared masked-panel pairs, zero shortage penalty. Negative values favour the LLM. O1*–O3* are rescaled original policies; M1–M3 are masked generations. The consistent signs support aggregate ranking stability on this panel, not independent replication across datasets or stability of individual policies.

340, 342 and 342 policies were scorable, and 5, 4 and 4 were rejected by the validity gate. Table 5 reports the replication.

Table 5: Six-run comparison on the intersection of 338 pairs feasible for all runs and seven comparators. Cost is mean holding-plus-ordering cost in masked units; fills and cost differences are percentages. O1*–O3* are rescaled frozen originals, M1–M3 fresh masked generations. Different scaling and pair inclusion explain differences from Table 3; six runs are not independent datasets.

Run	Cost	Mean fill	Aggr. fill	vs ERP	vs History	vs SBA	vs Empirical	vs Tuned
O1*	2,922	98.34	76.44	−11.4	+6.0	+8.3	+18.0	−10.6
O2*	2,925	98.11	76.27	−11.3	+6.2	+8.5	+18.2	−10.5
O3*	2,919	98.37	76.46	−11.5	+5.9	+8.2	+17.9	−10.7
M1	2,898	98.33	76.34	−12.2	+5.1	+7.4	+17.0	−11.3
M2	2,932	98.37	76.38	−11.1	+6.4	+8.7	+18.4	−10.3
M3	2,892	98.37	76.29	−12.3	+4.9	+7.2	+16.8	−11.5

On each masked run's own feasible cohort, the LLM is 11–12% cheaper than ERP-derived parameters and 5–6%, 7–9% and 16–18% costlier than History-only, Croston/SBA and empirical-quantile code. Matched costs relative to rescaled original run 1 are 0.991, 1.004 and 0.991, not comparisons with confidential unscaled means. Table 5 and Figures 2–4 instead use the same 338 pairs across all six runs, avoiding changes of denominator. Figure 3 confirms the PL3 reversal. Masking changes portfolio weights and rounding, so this tests

robustness of rankings under transformation, not equality of original and masked numerical results.

6.10. Sensitivity analyses

Plant C calendar.. Plant C's working-day flag required reconstruction, so we re-scored all policies under four alternatives (any-plant, strict-all, exclusion of Plant C, calendar-day arrivals). Where Plant C can be scored, the LLM remains 31–35% more expensive than SAP-derived there, so the Plant C result is not an artefact of the repair.

Shortage penalty.. Rescoring the original fixed trajectories at one and five times unit cost makes the simulation-tuned policy about 33% cheaper than SAP-derived; LLM run 1 costs 2.4% more than SAP-derived at both multiples. These are accounting sensitivities, not policy optimization at each penalty. Table 6 retains the original-scale analysis; the public six-run evidence also records shortage exposure for masked-scale rescoring.

Break-even shortage price.. Table 6 re-scores the same replays for a lost-sales penalty between zero and one unit cost per unit. The LLM is cheaper than SAP-derived parameters only while the penalty stays below 0.9% of the unit cost, and cheaper than the History-only and empirical-quantile code only above 1.5% and 2.4%, by about 1% or less. It is cheaper than the SLT-informed and ERP-floor ORs at every penalty and cheaper than the simulation-tuned policy only below 0.06%. From a penalty of 1% of the unit cost, the SAP-derived, History-only, Croston/SBA, empirical-quantile and LLM policies lie within 4% of each other, while the simulation-tuned policy is 28–33% cheaper than SAP-derived parameters. The ranking between the LLM and the formulas is thus decided by a shortage price that the primary basis sets to zero, which is why it is reported as a sensitivity and not as a headline result.

Table 6: Break-even shortage price. Mean total cost per pair (holding, ordering and shortage cost) in the hard-limit replay when each lost unit is charged at the given multiple of the unit cost, and the break-even multiple at which the LLM and the arm have equal mean cost (common-feasible cohort, $n = 346$).

Arm	0	0.01	0.1	1	Break-even multiple of unit cost
SAP-derived	3,305	19,613	166,388	1,634,136	0.0089; LLM cheaper below
History-only OR	2,764	19,609	171,207	1,687,191	0.0145; LLM cheaper above
Croston/SBA OR	2,705	19,427	169,927	1,674,925	none; LLM costlier on the grid
Empirical-quantile OR	2,489	19,402	171,618	1,693,784	0.0242; LLM cheaper above
Simulation-tuned	3,266	14,174	112,344	1,094,043	0.0006; LLM cheaper below
LLM (mean of three runs)	2,929	19,660	170,238	1,676,014	–

7. Discussion, Limitations and Research Perspectives

What the evidence shows.. A prompt-configured LLM generates checkable (r, Q) candidates: all original proposals passed the static gate, but masked generations include missing and invalid outputs. On scored cohorts they cost less than ERP-derived parameters and more than three history-based policies at zero shortage penalty. Policy similarity, a reorder-point uplift and run-to-run variation help characterize this gap; they do not establish the LLM's internal calculations or an inherent inability to outperform code. Information-matched inputs need not imply identical use of those inputs. InventOR is a reviewed candidate generator, not evidence of optimization superiority. Written rationales and escalation flags could aid review, but development effort, explanation quality, planner time and realized benefits were not measured.

Implications for practice.. (1) Benchmark against data-driven code, not ERP values alone; zero safety stocks and large safety lead times make input treatment important. (2) Price shortages before interpreting cost rankings, which otherwise omit service losses (Section 6.10). (3) Verify trajectory storage and order-level MOQ jointly: static-compliant policies exceeded capacity, while clipped replay can violate MOQ. (4) Expect plant differences and define review and escalation rules before use [11]. (5) Control prompt versions, retain call provenance and repeat generations. (6) Generation took minutes per material rather than the much faster formula calculations; no controlled development-effort

or planner-time study was conducted. Periodic regeneration is a possible planning use [88], not a demonstrated operational benefit.

Limitations.. (i) One manufacturer and three plants limit generalization; six generations reuse one underlying panel, not six independent datasets. (ii) Model identity is operator-declared; runtime identifier and sampling settings are unavailable. (iii) No planner outcomes or explanation-quality measures were collected. (iv) Deterministic mean-lead-time arrivals, empty starting pipelines and ignored observed receipts simplify operations; daily order review and capacity clipping can violate supplier MOQ. (v) Zero shortage penalty is the primary basis, not an equal-service comparison. Positive penalties rescore fixed trajectories rather than re-optimize policies. Demand concentration and within-plant dependence limit bootstrap interpretation. (vi) Only single-item, single-echelon (r, Q) policies are compared. (vii) No trained neural or reinforcement-learning baseline is included; the tuned arm's in-sample initialization differs from evaluation. ERP-derived policies use ERP safety stock and safety lead time with our order quantity, not observed practice. (viii) The public masked panel and frozen policies support offline scoring, not exact regeneration of confidential original results or independent reproduction of proprietary-model inference. Per-pair masking changes portfolio weights and rounded quantities.

Research perspectives.. (1) *Ensembling and hybrids:* aggregating several generations may reduce individual-policy noise [35], and combining the best data-driven policy with LLM review [89] could capture both the OR cost level and the LLM's flags. (2) *Comparison with trained methods:* benchmarks against reinforcement-learning policies [44, 38, 39] and richer policy families such as (s, S) [66, 67] on identical data would clarify the trade-off between setup effort and policy quality. (3) *Planner-in-the-loop field trials:* prospective studies should compare planners with and without the aid on matched materials, measuring planner time,

override rates and realized service and inventory, and examine adoption drivers [23, 22, 6].

8. Conclusion

On 365 industrial plant-material pairs, three original generations and three masked-release generations show that a prompt-configured LLM can propose replenishment parameters without project-specific training, but not invariably without invalid or missing outputs. In the original common-feasible cohort it costs about 11% less than ERP-derived parameters at slightly lower mean service, yet 6%, 8% and 18% more than history-only, Croston/SBA and empirical-quantile policies. Masked generations preserve this ranking, not independent external validity. Pricing shortages changes the ordering of fixed policies; static storage checks do not ensure trajectory feasibility, and capacity-clipped orders need not satisfy MOQ. Parameter similarity to textbook code is descriptive, not proof of internal formula fidelity. InventOR is supported as a checkable candidate generator inside a proposed reviewed process, not as an optimizer that beats data-driven baselines. Prospective planner studies and evaluation under simultaneous order-size and capacity constraints remain necessary.

Appendix A. Formal Definitions

Reference calculation.. These are standard textbook calculations [72, 82]. For pair i, $\bar{d}_i, \sigma_{D,i}$ describe pre-cutoff working-day demand; $L_i, \sigma_{L,i}$ describe recorded lead times; K_i is ordering cost, h_i annual holding cost per unit, and z_i the normal service-target quantile. Set $Q_i = \max\{\mathrm{MOQ}_i, \lceil\sqrt{2K_i \cdot 250\bar{d}_i/h_i}\rceil\}$, $\sigma_{LTD,i} = \sqrt{L_i^{\mathrm{eff}}\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, $SS_i = \lceil z_i\sigma_{LTD,i}\rceil$ and $R_i = \lceil\bar{d}_i L_i^{\mathrm{eff}} + SS_i\rceil$. Here $L_i^{\mathrm{eff}} = L_i$ for History-only and $L_i + \max(\text{ERP SLT}_i, 0)$ for SLT-informed and ERP-floor; the latter also floors SS_i at ERP safety stock. SAP-derived uses ERP safety stock, the same Q_i, and ERP safety lead time in its reorder point. Projection enforces $SS_i + Q_i \le \texttt{max_storage_units}_i$ when MOQ plus safety stock fits.

Simulation.. On each post-cutoff working day, due receipts arrive first, demand is served, unmet demand is lost, and at most one order is placed when inventory position (on hand plus on order) is at or below R_i. It arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. Cost is $TC_i = h_i \bar{I}_i T/250 + K_i N_i$, where $\bar{I}_i$ is mean on-hand stock over T working days and N_i counts orders. Fill is served divided by demanded units. Hard-limit orders are $\min\{Q_i, \text{capacity}_i - \lfloor \text{position}_i \rfloor\}$ if positive, even when below MOQ; no order is placed otherwise. Observed opening stock is not clipped.

Author Contributions

Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review and editing: Aris Dressino.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

Raw operational data are confidential. The company approved a masked panel of the same 233,078 records, with pseudonymous identifiers and secret per-pair

quantity and price factors. The panel, rescaled frozen policies, three masked generations, sanitized provenance, scoring code, aggregate evidence and manuscript sources are at `github.com/DressPD/inventor_public`. The archive concept DOI is `https://doi.org/10.5281/zenodo.22262051`; at the audit date (24 October 2026), its latest archived version is v1.2.0, not the current masked release. Supplementary File S1 contains the prompt. Runner-hash amendments are recorded separately from original call provenance, with retained original hashes and amendment histories; they must not be read as original execution hashes. Model identity (Claude Sonnet 4.6) is operator-declared; runtime identifier and sampling settings are unavailable.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

OpenCode AI-assisted tools, including GPT-6.1 Sol for the final audit, assisted language editing, restructuring, consistency and citation checks, LaTeX preparation and analysis-script development. These preparation tools are distinct from the operator-declared Claude Sonnet 4.6 deployment evaluated in the study. The author is responsible for verifying the complete tool history, reviewing the resulting text and code, and the content of the final article. Figures are deterministic diagrams and data plots rendered from source files, not image-model outputs.

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